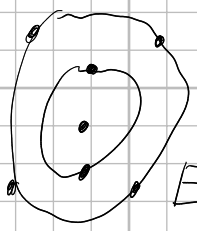
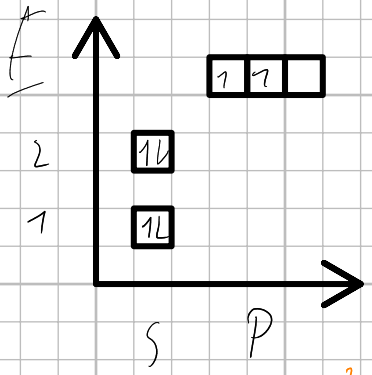
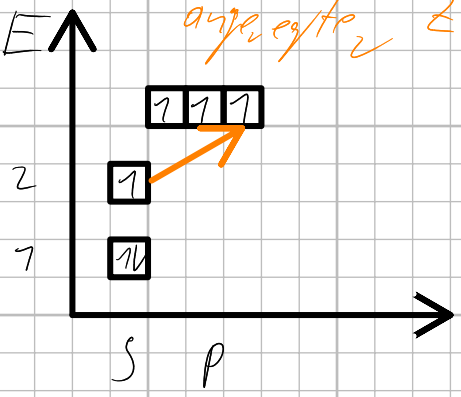
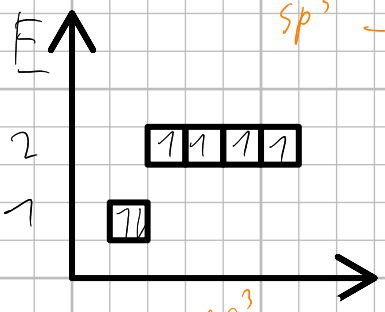
6 C 6 e⁻







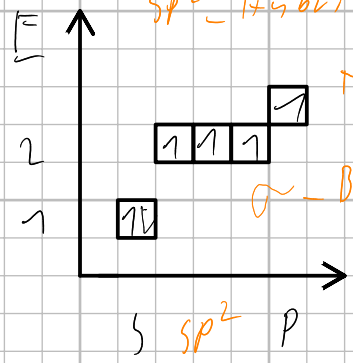


sp³ – Hybridisierung

2 C (schwarz)
6 H (weiß)
3 lange Vb.
7 kurze Vb.

METHAN CH_4
ETHAN C_2H_6

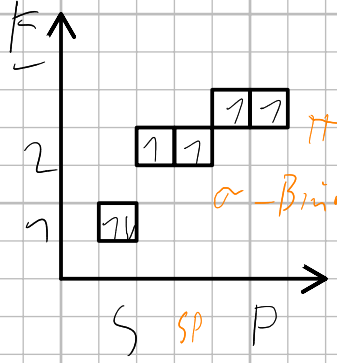
σ-Bindung:
Einfachbindung, frei drehbar, reaktionsträge, tetraedrisch



Bsp. ETHEN C_2H_4

π-Bindung: Nicht frei drehbar
reaktionsfreudig

Doppelbindung:
1 σ-Bindung, 1 π-Bindung



Bsp. ETHIN, C_2H_2

linear
Dreifachbindung:
1 σ-Bindung, 2 π-Bindungen

Kohlenwasserstoffe

- kettenförmig → **ALIPHATEN** (acyclisch)
  - gesättigt → **ALKANE** (Einfachbindungen)
  - ungesättigt → **ALKENE** (min. 1 Doppelbindung)
  - ungesättigt → **ALKINE** (min. 1 Dreifachbindung)
- ringförmig
  - alicyclisch
  - → **AROMATEN**

Bsp. Ethan

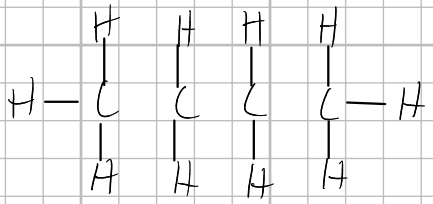
Ethen

Ethin



Skelettschreibweise

Bsp Butan



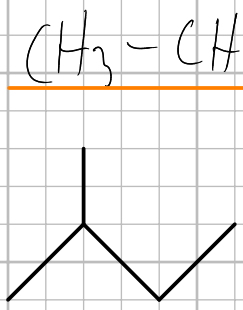
Strukturformel

$CH_3-CH_2-CH_2-CH_3$

Nonan

$CH_3-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_2-CH_3$

ALKANE



$CH_3-CH_2-CH(CH_3)-CH_3$ — Methyl

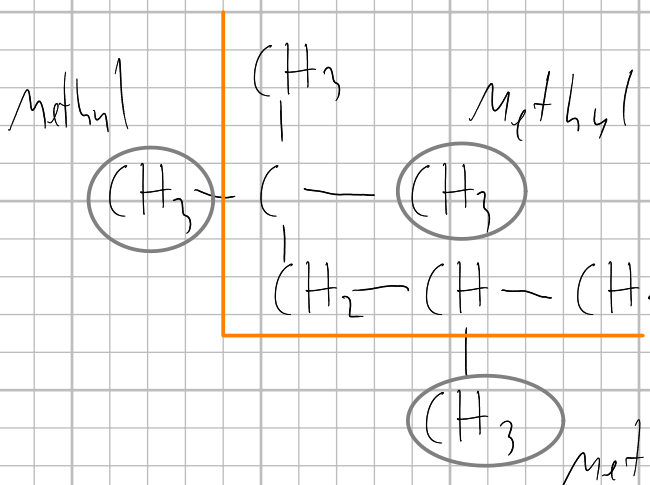
2-Methylbutan

$CH_3-CH(CH_3)-CH_2-CH_3$

$CH_3-CH(CH_3)-CH(CH_2-CH_3)-CH_2-CH_2-CH_2-CH_3$

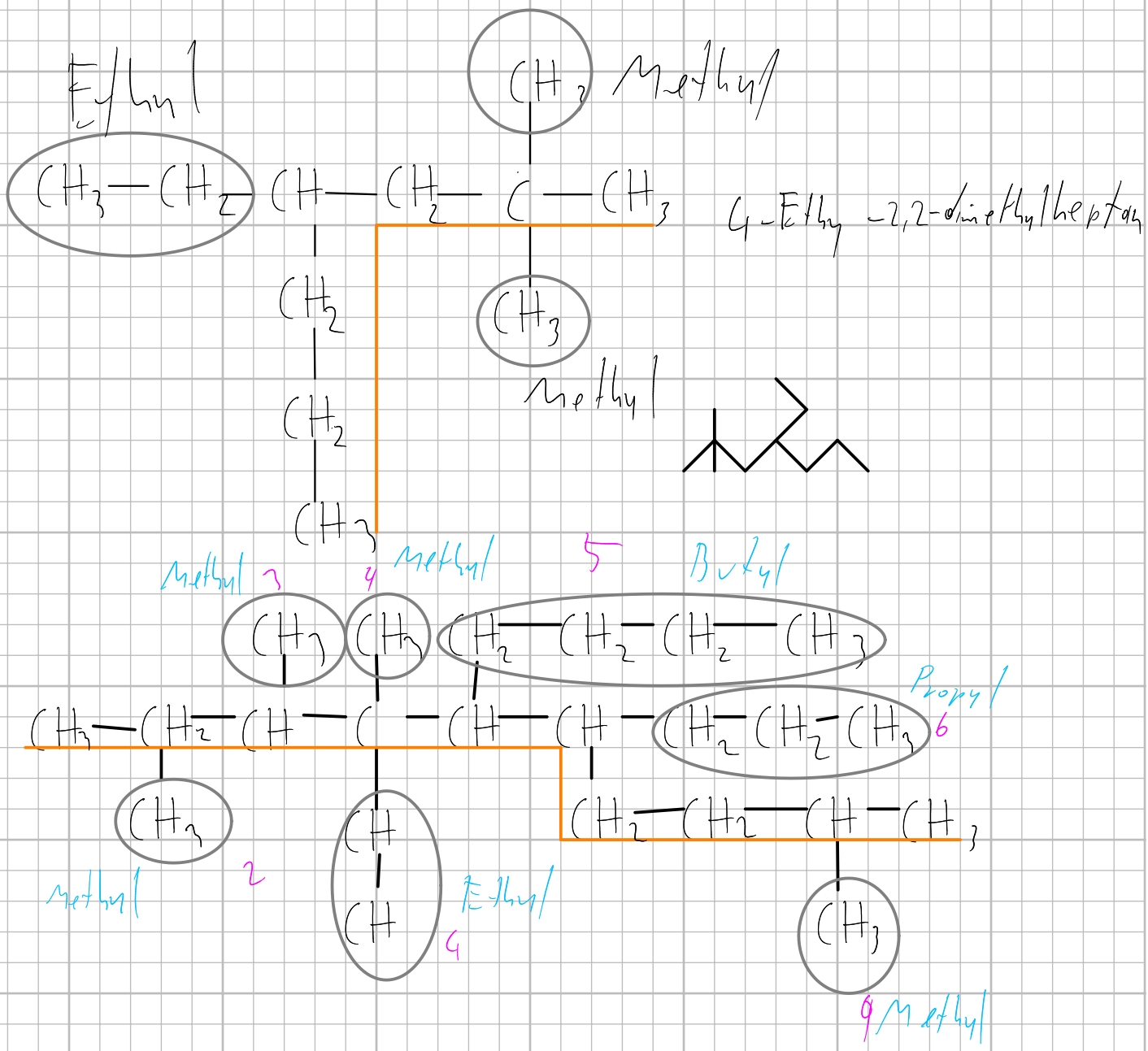
Methyl — Ethyl

3-Ethyl-2-methylheptan



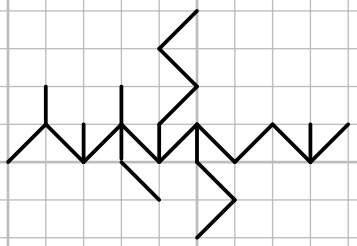
Methyl — Methyl — Methyl

2,2,4-Trimethylpentan



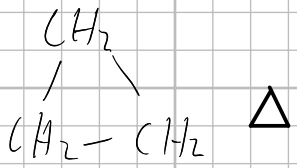
4-Ethyl-2,2-dimethylheptan

5-Butyl-4-ethyl-2,3,4,9-tetramethyl-6-propyldecan

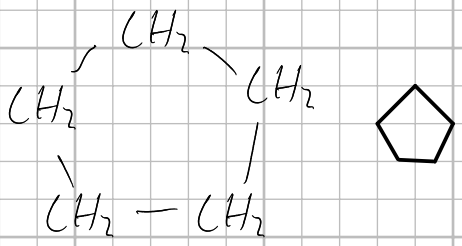


alicylische Kohlenwasserstoffe

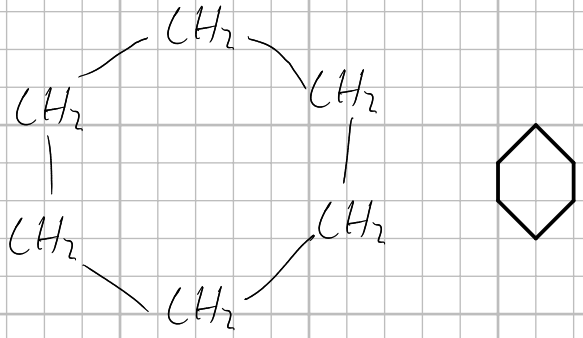
Cyclopropan



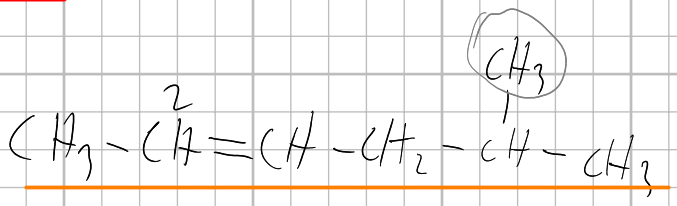
Cyclopentan



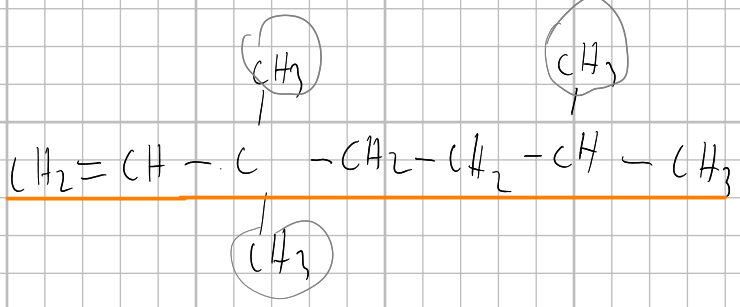
Cyclohexan



Alkene

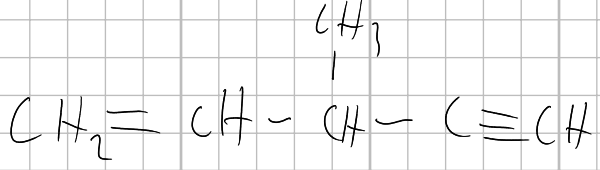


5-Methylhex-2-en



3,3,5-Trimethylhept-1-en

4) 3-Methylpent-1-en-4-in





1) 2,5-Dimethylhex-3-in

